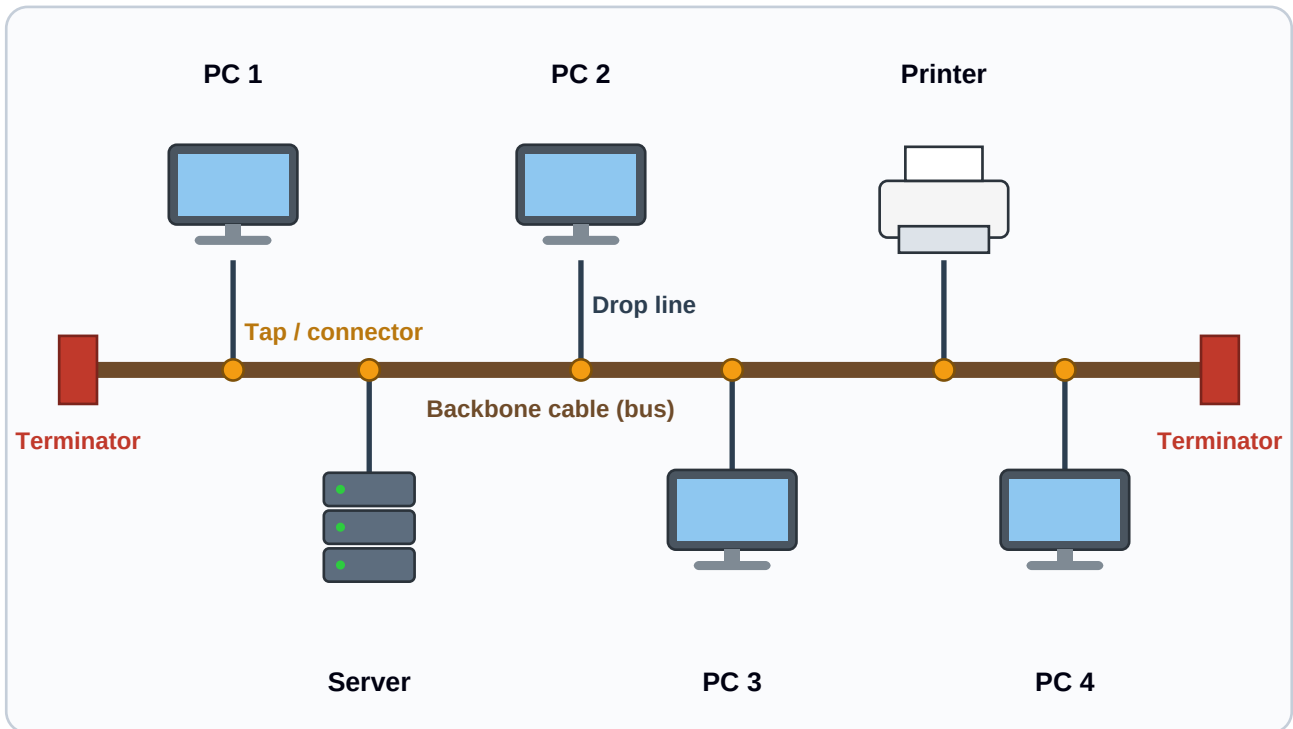


Assignment: Day 7

Network Topologies

Q1. Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.

Ans:



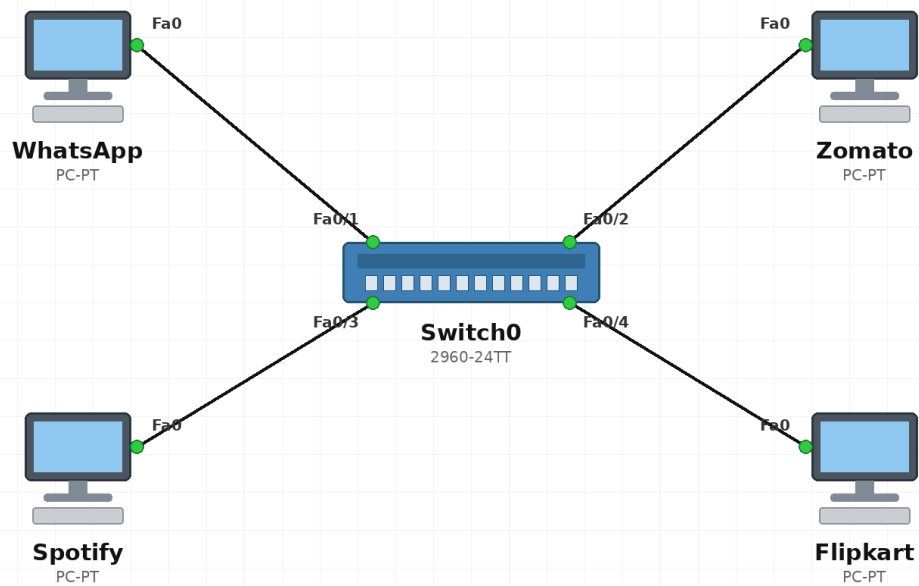
Real-world example: Early Ethernet networks (10BASE2) in small offices and computer labs connected all computers to a single coaxial cable. Modern vehicles also use the CAN bus, where many electronic units share one common line.

Q2. In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app and take a screenshot of your topology.

Ans: Star topology with one switch (Switch0) and four PCs named WhatsApp, Zomato, Spotify and Flipkart. Each PC is connected to a separate switch port using a copper straight-through cable.

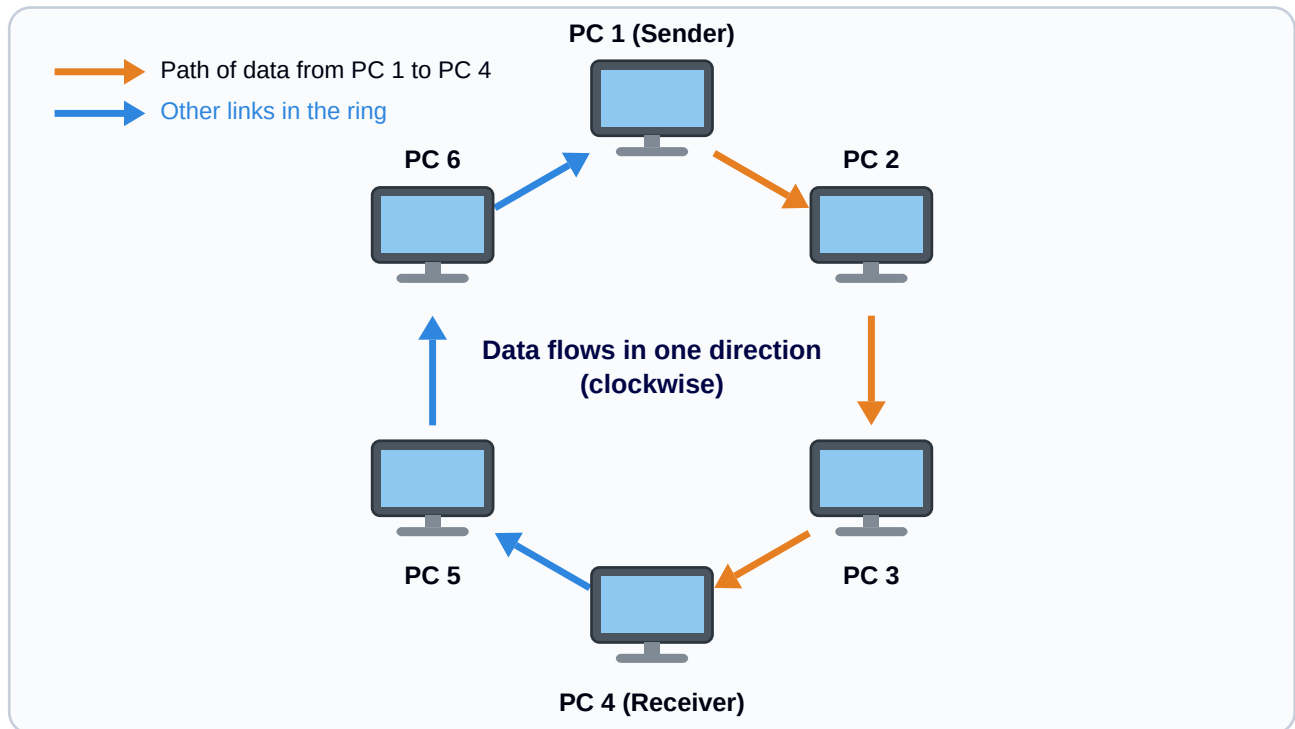
Logical | Star topology: 1 switch, 4 PCs

Realtime



Q3. Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.

Ans:



In a ring topology every device is connected to exactly two neighbours, forming a closed loop, and data travels in only one direction around it. When PC 1 sends data to PC 4, the data passes through PC 2 and PC 3, and each device checks the destination address and forwards the data to the next device if it is not meant for it. PC 4 finally accepts the data, and the ring keeps working as a continuous path.

Q4. Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

Ans:

Topology	Advantages	Disadvantages
Star	1. If one cable or device fails, only that device is affected and the rest of the network keeps working. 2. Easy to install, troubleshoot, and to add or remove devices, because everything is managed from the central switch.	1. Single point of failure: if the central switch fails, the whole network stops. 2. Needs more cable, because every device needs its own cable to the switch, and the switch adds cost.
Mesh	1. Very reliable: there are many paths between devices, so data can be re-routed if a link fails. 2. Dedicated links give high speed and better privacy, with no traffic sharing or collisions.	1. Very expensive: a full mesh of n devices needs $n(n-1)/2$ links, for example 10 links for 5 devices. 2. Complex to install and maintain, because each device needs many ports and cables.